

# Lecture 01: Electron as a Particle

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# History of electrons as particle-like entities

- ▶ Late 1800s: electrons discovered as discrete charged particles with definite mass
- ▶ Classical expectation: electrons travel in straight lines, scatter like billiard balls
- ▶ One can try to use this idea to explain a phenomena related to electrodynamics

## Examples of phenomena of interest

- ▶ Electrical conductivity of materials
- ▶ Magnetic behavior of materials
- ▶ Optical behavior of metals
- ▶ Electromagnetic waves
- ▶ ...

*If electrons are particles, their behavior should be entirely predictable from classical mechanics*

**Electron as a particle:** can we explain the behavior of electrical conductivity in metals by assuming that electrons are particles?

# The Drude model: electrons as classical particles

- ▶ Drude (1900): treat conduction electrons as a classical gas of free particles
- ▶ Electrons move randomly, colliding with fixed ion cores at average interval  $\tau$
- ▶ Between collisions, electrons respond freely to applied electric fields
- ▶ Model explains conductivity, Hall effect, and thermal transport qualitatively



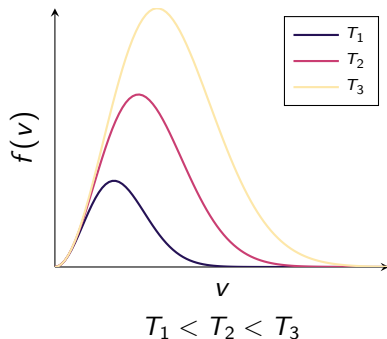
Random thermal motion

# Maxwell-Boltzmann velocity distribution

- ▶ Classical gas: velocities follow Maxwell-Boltzmann distribution at temperature  $T$
- ▶ Distribution is isotropic (no preferred direction)
- ▶  $\langle \mathbf{v} \rangle = 0$  but  $\langle |\mathbf{v}| \rangle \neq 0$
- ▶ Higher  $T \rightarrow$  broader distribution

**Distribution:**

$$f(v) = 4\pi n \left( \frac{m}{2\pi k_B T} \right)^{3/2} v^2 \exp \left( -\frac{mv^2}{2k_B T} \right)$$



# Thermal velocity of electrons

## Derivation from equipartition theorem:

- ▶ Each translational degree of freedom has energy  $\frac{1}{2}k_B T$
- ▶ Three dimensions  $\rightarrow$  total kinetic energy  $= \frac{3}{2}k_B T$  per electron

$$\frac{1}{2}mv_{th}^2 = \frac{3}{2}k_B T \quad \Rightarrow$$

$$v_{th} = \sqrt{\frac{3k_B T}{m}}$$

## Numerical estimate at $T = 300 \text{ K}$ :

- ▶  $v_{th} \approx 1.2 \times 10^5 \text{ m/s}$  for free electrons
- ▶ Thermal velocity sets the scale for electron motion between collisions

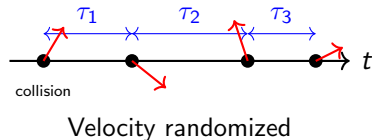
*Notation:*  $m$  = electron mass,  $k_B$  = Boltzmann constant,  $T$  = temperature

# Collision time and mean free path

- **Collision time**  $\tau$ : average time between electron-ion scattering events
- **Mean free path:**

$$\ell = v_{th} \cdot \tau$$

- Typical metals:  $\tau \approx 10^{-14}$  s,  $\ell \approx 1\text{--}10$  nm
- After collision: velocity randomized  $\rightarrow$  memory erased



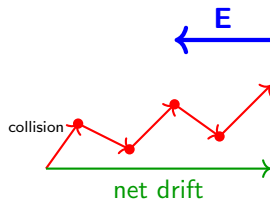
# Electron motion under applied electric field

- ▶ Applied field  $\mathbf{E}$  accelerates electrons:

$$\mathbf{F} = -e\mathbf{E}$$

(opposite to field direction)

- ▶ Between collisions, electron gains velocity  $\Delta v = -eEt/m$
- ▶ Collisions randomize velocity, resetting drift to zero
- ▶ Average time between collisions defines  $\tau$





## Derivation of drift velocity

**Key idea:** Average velocity gained before next collision

- ▶ Acceleration:  $a = -eE/m$
- ▶ Average time to next collision:  $\tau$
- ▶ Average velocity gained:

$$v_D = -\frac{eE\tau}{m}$$

**Important observations:**

- ▶ Drift velocity  $v_D \ll v_{th}$  (thermal velocity)
- ▶ Typical values:  $v_{th} \sim 10^5$  m/s,  $v_D \sim 10^{-3}$  m/s
- ▶ Negative sign: electrons drift *opposite* to **E**

*Notation:*  $e$  = elementary charge (positive),  $\tau$  = collision time

## From drift velocity to conductivity $\sigma$

### Derivation of Ohm's law:

- ▶ Current density = charge density  $\times$  velocity:

$$J = -N_e e v_D$$

- ▶ Substitute  $v_D = -eE\tau/m$ :

$$J = -N_e e \left( -\frac{eE\tau}{m} \right) = \frac{N_e e^2 \tau}{m} E$$

**Ohm's law:**  $J = \sigma E$  identifies:

$$\sigma = \frac{N_e e^2 \tau}{m}$$

*Notation:*  $N_e$  = electron density,  $\sigma$  = electrical conductivity

## Mobility $\mu$ as intrinsic property of materials

**Definition:** Drift velocity per unit electric field

$$\mu = \frac{e\tau}{m} \quad \Rightarrow \quad v_D = -\mu E$$

**Conductivity as a function of mobility:**

$$\sigma = N_e e \mu$$

- ▶ Mobility reflects scattering mechanisms (independent of  $N_e$ )
- ▶ Units:  $\mu$  in  $\text{m}^2/(\text{V}\cdot\text{s})$  or  $\text{cm}^2/(\text{V}\cdot\text{s})$
- ▶ Separates material quality ( $\mu$ ) from doping level ( $N_e$ )

# Why semiconductors are different from metals

- ▶ **Metals:** high  $N_e$  ( $\sim 10^{28} \text{ m}^{-3}$ ), moderate  $\mu$   
→ high  $\sigma$  ( $\sim 10^7 \text{ S/m}$ )
- ▶ **Semiconductors:** low  $N_e$  ( $\sim 10^{16} \text{ m}^{-3}$ ), high  $\mu$   
→ low  $\sigma$  ( $\sim 10^{-3} \text{ S/m}$ )
- ▶ Temperature dependence is different

$$T \uparrow \Rightarrow \mu_{\text{metal}} \downarrow$$

$$\Rightarrow \sigma_{\text{metal}} \downarrow$$

$$T \uparrow \Rightarrow N_e \uparrow$$

$$\Rightarrow \sigma_{\text{semic}} \uparrow$$

**Electron as a particle:** can we explain the behavior of electrons under applied magnetic fields if we assume they are particles?

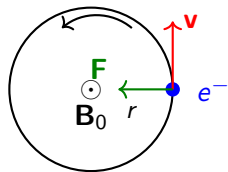
# Electron motion in a magnetic field

**Notation:**  $\mathbf{B}_0$  = applied magnetic field,  $\mathbf{v}$  = electron velocity,  $r$  = orbit radius

Lorentz force

$$m \frac{d\mathbf{v}}{dt} = -e\mathbf{v} \times \mathbf{B}_0 \quad (1)$$

- ▶ Force  $\mathbf{F} = -e\mathbf{v} \times \mathbf{B}_0$  acts  $\perp$  to velocity
- ▶ No work done: kinetic energy conserved
- ▶ Electron executes **circular orbit**
- ▶ Orbit radius:  $r = mv/(eB_0)$



## Cyclotron frequency

**Goal:** Find angular frequency of circular electron orbit

1. **Centripetal acceleration:**  $a = \frac{v^2}{r}$  for circular motion
2. **Equate Lorentz force to centripetal force:**

$$evB_0 = \frac{mv^2}{r} \quad (2)$$

3. **Solve for radius:**  $r = \frac{mv}{eB_0}$
4. **Angular frequency:**  $\omega = \frac{v}{r} = \frac{v \cdot eB_0}{mv}$

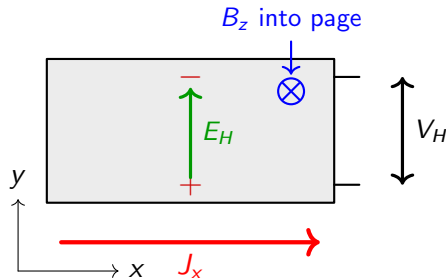
## Cyclotron frequency

$$\boxed{\omega_c = \frac{eB_0}{m}} \quad (3)$$

## Hall effect: geometry and setup

- ▶ Current density  $J_x$  flows along  $+x$
- ▶ Magnetic field  $B_z$  points *into the page* ( $-z$ )
- ▶ Lorentz force deflects carriers ( $\mathbf{F} = q\mathbf{v} \times \mathbf{B}$ )
- ▶ Steady state: transverse Hall field  $E_H$  builds up

*Result:* a transverse Hall voltage  $V_H$  appears between the top and bottom edges





## Deriving the Hall coefficient

**Notation:**  $\mathbf{F}$  = force,  $e$  = elementary charge,  $\mathbf{E}$  = electric field,  $\mathbf{v}$  = velocity,  $\mathbf{B}$  = magnetic field,  $v_D$  = drift velocity,  $J_x$  = current density,  $N_e$  = electron density

- ▶ Lorentz force on electron:  $\mathbf{F} = -e(\mathbf{E} + \mathbf{v} \times \mathbf{B})$
- ▶ In steady state, transverse current  $J_y = 0$  requires force balance
- ▶ Hall field  $E_H$  develops to cancel magnetic deflection
- ▶ Balance condition:  $eE_H = ev_D \cdot B$  where  $v_D$  is drift velocity

**Force balance condition:**

$$eE_H = ev_D B = \frac{J_x B}{N_e} \quad (4)$$

(Using  $J_x = N_e e v_D$ )

## Hall Coefficient: final result

**Notation:**  $R_H$  = Hall coefficient,  $E_H$  = Hall field,  $V_H$  = Hall voltage,  $w$  = sample width,  $t$  = sample thickness,  $I$  = current

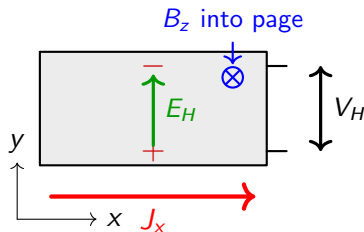
- ▶ Rearranging:  $E_H = J_x B / (N_e e)$  defines Hall coefficient  $R_H$
- ▶  $R_H = 1 / (N_e e)$  depends only on carrier density, not mobility or mass
- ▶ Sign of  $R_H$  reveals carrier type: negative for electrons
- ▶ Measurement:  $V_H = E_H \cdot w = R_H \cdot I \cdot B / t$

**Hall coefficient:**

$$R_H = \frac{E_H}{J_x B} = \frac{1}{N_e e} \quad (5)$$

# Experimental carrier characterization

- ▶ Measure Hall voltage  $V_H$  at known current  $I$  and field  $B$
- ▶ Calculate  $R_H = V_H \cdot t / (I \cdot B)$  directly
- ▶ Carrier density:  $N_e = 1 / (R_H \cdot e)$   
(typically  $\sim 10^{28} \text{ m}^{-3}$  for metals)
- ▶ Combined with  $\sigma$ : extract mobility



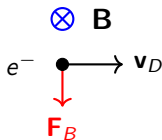
## Mobility from Hall measurement:

$$\mu = \sigma R_H = \frac{\sigma}{N_e e} \quad (6)$$

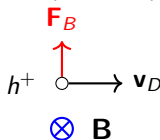
## Positive Hall coefficients

- ▶ Some metals (Be, Zn, Cd) show positive  $R_H$  — “wrong” sign?
- ▶  $R_H > 0$  implies carriers with positive charge: **holes** in valence band
- ▶ Hole = absence of electron; moves opposite to electron motion
- ▶ Full explanation requires band theory (we will discuss that in future lectures)

**Electrons** ( $R_H < 0$ )



**Holes** ( $R_H > 0$ )



**Electron as a particle:** can we explain the behavior of electromagnetic wave propagation in metals by assuming electrons are particles?

**Pause:** first, we have to remember the laws of electromagnetism

## Maxwell's equations in vacuum

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\varepsilon_0} \quad (\text{Gauss' Law})$$

$$\nabla \cdot \mathbf{B} = 0 \quad (\text{Gauss' Law for Magnetism})$$

$$\nabla \times \mathbf{E} = - \frac{\partial \mathbf{B}}{\partial t} \quad (\text{Faraday's Law})$$

$$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \varepsilon_0 \frac{\partial \mathbf{E}}{\partial t} \quad (\text{Ampère-Maxwell's Law})$$

With the definitions:  $\mathbf{E}$  = electric field,  $\mathbf{B}$  = magnetic density field,  $\rho$  = charge density,  $\varepsilon$  = dielectric permittivity,  $\mu$  = magnetic permittivity,

## Maxwell's equations (general)

Or with the definitions  $\mathbf{D} = \varepsilon \mathbf{E}$  ( $\mathbf{D}$  = displacement field) and  $\mathbf{B} = \mu \mathbf{H}$  ( $\mathbf{H}$  = magnetic field),

$$\nabla \cdot \mathbf{D} = \rho$$

$$\nabla \cdot \mathbf{B} = 0$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

## EM waves in conductors: the wave equation

**Notation:**  $\mathbf{H}$  = magnetic field intensity,  $\mathbf{J}$  = current density,  $\mathbf{D}$  = displacement field,  $\sigma$  = conductivity,  $\varepsilon$  = permittivity,  $\mu$  = permeability

- ▶ Maxwell's equations with conduction current:

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t} = \sigma \mathbf{E} + \varepsilon \frac{\partial \mathbf{E}}{\partial t}$$

- ▶ For good conductors at moderate frequencies:  $\sigma \gg \omega \varepsilon$   
(conduction dominates over displacement current)
- ▶ Wave equation becomes diffusion-like, not wave-like

**Diffusion-like behavior of electromagnetic waves in conductors:**

$$\nabla^2 \mathbf{E} = \mu \sigma \frac{\partial \mathbf{E}}{\partial t} \quad (7)$$



## Skin depth in conductors

**Notation:**  $k$  = complex wave vector,  $\delta$  = skin depth,  $\omega$  = angular frequency

- ▶ Assume plane wave  $\mathbf{E} = \mathbf{E}_0 \exp(ikz - i\omega t)$  propagating into conductor ( $z > 0$ )
- ▶ Substituting into wave equation:

$$k^2 = i\omega\mu\sigma \quad (8)$$

$$k = \frac{1+i}{\delta} \quad \text{where} \quad \boxed{\delta = \sqrt{\frac{2}{\omega\mu\sigma}}} \quad (9)$$

- ▶ Complex  $k$  gives both propagation and exponential decay
- ▶ Real part: oscillation; Imaginary part: attenuation

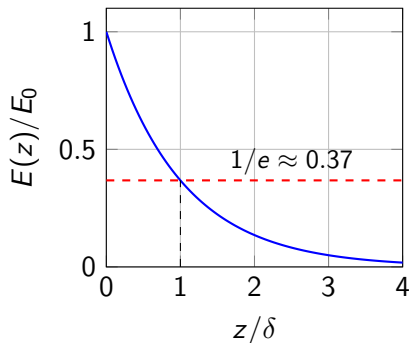
# Exponential decay of EM fields in conductors

- ▶ Field amplitude:

$$E(z) = E_0 e^{-z/\delta}$$

decays to  $1/e$  at depth  $\delta$

- ▶ Power density decays as  $e^{-2z/\delta}$ :  
energy confined to thin surface layer
- ▶ Higher frequency  $\rightarrow$  smaller  $\delta \rightarrow$  fields  
confined closer to surface



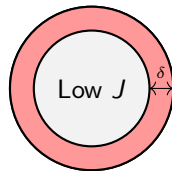
# High-frequency resistance and AC losses in conductors

- ▶ Current flows only in skin layer of thickness  $\delta$ : effective cross-section reduced
- ▶ AC resistance increases as

$$R_{AC} \propto \sqrt{\omega}$$

compared to DC resistance

- ▶ Copper at 1 MHz:  $\delta \approx 66 \mu\text{m}$
  - ▶ Copper at 1 GHz:  $\delta \approx 2 \mu\text{m}$
- ⇒ Use hollow conductors at high frequency!



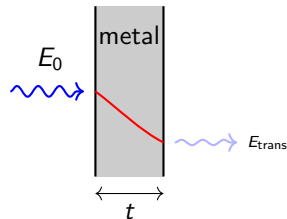
Current in skin layer

# Electromagnetic shielding in conductors

- ▶ Shielding effectiveness is proportional to the thickness  $t$ :

$$SE \propto (t/\delta) \text{ dB}$$

- ▶ **Example:** 1 mm Cu at 1 MHz:  
 $t/\delta \approx 15 \Rightarrow SE \approx 130 \text{ dB}$   
(excellent shielding!)



## Plasma frequency

**Notation:**  $\omega_p$  = plasma frequency,  $N_e$  = electron density,  $e$  = elementary charge,  $m$  = electron mass,  $\varepsilon$  = dielectric permittivity

- ▶ Plasma frequency  $\omega_p$  marks transition between **opaque** and **transparent** regimes
- ▶ Arises from collective oscillation of free electrons against fixed ion background
- ▶ Depends only on electron density — intrinsic material property

**Plasma frequency:**

$$\boxed{\omega_p = \sqrt{\frac{N_e e^2}{m \varepsilon}}} = \sqrt{\frac{\sigma}{\varepsilon \tau}} \quad (10)$$

| $\omega < \omega_p$                      | $\omega > \omega_p$                  |
|------------------------------------------|--------------------------------------|
| Metal opaque<br>Waves reflected/absorbed | Metal transparent<br>Waves propagate |

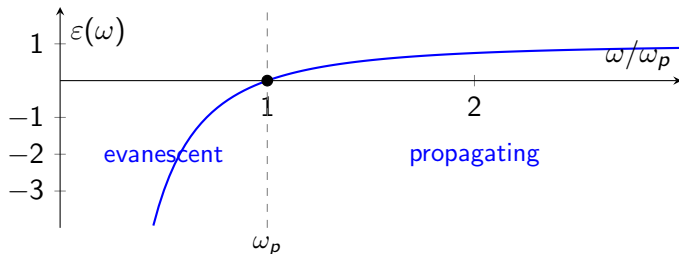
# Dielectric constant and wave propagation

**Notation:**  $\varepsilon(\omega)$  = effective dielectric constant,  $\omega_p$  = plasma frequency,  $k$  = wave vector

## Effective dielectric constant

$$\varepsilon_{\text{eff}}(\omega) = 1 - \frac{\omega_p^2}{\omega^2} \quad (11)$$

- ▶ For  $\omega < \omega_p$ :  $\varepsilon < 0$ ,  $k$  is imaginary  $\rightarrow$  **evanescent wave**
- ▶ For  $\omega > \omega_p$ :  $\varepsilon > 0$ ,  $k$  is real  $\rightarrow$  **propagating wave**



## Metallic reflectivity across the EM spectrum

- ▶ Metals **reflect** EM waves when  $\omega < \omega_p$  (visible light for most metals)
- ▶ Metals become **transparent** when  $\omega > \omega_p$  (UV regime)
- ▶ Typical metal:  $\omega_p \approx 2.2 \times 10^{15}$  Hz corresponds to UV wavelength  $\sim 140$  nm

